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DATABASE & CLOUD

- Oracle APEX, SQL, PL/SQL
- Postgres, SQLite, MongoDB
- Supabase, Prisma, Drizzle
- Docker, CI/CD, cloud platforms

FULL STACK & AI

- React, Next.js, TypeScript
- Node.js, Python, FastAPI
- Claude / OpenAI integrations
- Agentic workflows and support AI

MOBILE & EMBEDDED

- Flutter / Dart / iOS / Android
- C, C++, C#, Java
- BLE, CAN, OBD2, telemetry
- Firmware and bootloader workflows

DOMAIN

- Industrial machinery software
- Automotive tuning and racing
- Manufacturing operations
- Scientific / regulated systems

Matthew Phillips

Engineering Lead / Full-Stack Software Engineer

AI | Oracle APEX | Mobile | Embedded | Industrial Platforms

github.com/Stericallyhindered/PORTFOLIO

Summary

Engineering lead and full-stack, firmware, and embedded systems engineer with experience across Oracle APEX-backed database applications, AI automation, backend/API systems, industrial CAD/CAM and machine-control interfaces, cross-platform applications, microcontroller firmware, automotive tuning systems, telematics, hardware validation, and regulated analytical testing environments.

I have led development teams through the full lifecycle of connected products and business platforms: translating unclear requirements into buildable work, coordinating across software/hardware/support needs, reviewing implementation details, documenting decisions, mentoring junior developers, and shipping systems used by real customers and staff.

My strongest work is the connective tissue between systems: embedded firmware, bootloaders, remote application-to-device updates, Python test automation, CAD/CAM and operator interfaces, mobile and desktop applications, BLE/CAN/OBD2/UART communication, high-volume telemetry, cloud APIs, databases, AI support flows, and production tooling that connects physical products to customers, technicians, and business operations.

Leadership Profile

- Led five developers at Burger Motorsports through the development of multiple automotive hardware/software products, including JB4Pro, CANFlex, tuning devices, fuel injector controllers, cross-platform mobile applications, firmware-related workflows, support systems, and customer-facing tooling.
- Currently lead and mentor three junior developers at Stealth Machine Tools / Stealth Batteries while building AI integrations, CRM systems, customer support automation, internal dashboards, field/support apps, and business workflow platforms.
- Train one developer to take over ongoing product ownership as the current product lifecycle reaches a close, emphasizing documentation, maintainability, scoped changes, clean handoff, and production support.
- Comfortable acting as the technical bridge between owners, hardware engineers, firmware contributors, support staff, junior developers, and customers.
- Strong fit for small teams, ambiguous requirements, internal tooling, developer enablement, automation platforms, and practical AI systems where the goal is to make other people faster and less blocked.

Core Strengths

- Full-stack product engineering: React, Next.js, TypeScript, JavaScript, Node.js, Express, Python, FastAPI, REST APIs, WebSockets, auth, permissions, dashboards, admin tools, customer portals, and deployment documentation.
- Backend and integration work: API integrations, webhooks, provider adapters, notification flows, Stripe/payment workflows, shipping workflows, ticketing, logging, audit trails, role-aware state machines, structured operational data, and Oracle APEX-connected business databases.
- Oracle APEX and database application engineering: Oracle APEX, SQL, PL/SQL-style workflow thinking, customer/warranty/automation databases, reporting, production support, documentation, testing, and enhancements.
- AI systems: Claude/Anthropic and OpenAI-style integrations, AI support agents, chat workflows, document/context ingestion, ticket triage, warranty/claims intake, onboarding, knowledge-base lookup, analytics, escalation paths, and human handoff flows.
- Internal tools and workflow platforms: CRM/order systems, developer/support tooling, operational dashboards, approval workflows, customer records, machine/device records, sales/dealer portals, affiliate workflows, field-service tools, and support history.
- Industrial machine and CAD/CAM software: full-stack operator and technician interfaces for fiber laser welders and cutters, plasma CNC equipment, press brakes, benders, and related machine tools; CAD/CAM workflows, job and machine configuration, diagnostics, telemetry, field support, and remote service tooling.
- Mobile and hardware-connected software: Flutter/Dart applications across Windows, macOS, iOS, and Android; BLE, UART/serial, CAN bus, OBD2, TCP/IP, device discovery, live telemetry, diagnostics, datalogging, settings, calibration, and application-to-device control.
- Firmware and embedded systems: C/C++, microcontrollers and embedded development platforms, ESP32/ESP-IDF, Arduino, Raspberry Pi-class systems, device drivers, interrupts, state machines, watchdogs, persistent configuration, bootloaders, remote firmware delivery, HEX parsing, block erase/write, checksum validation, and recovery behavior.
- Firmware validation and test engineering: Python automated test scripts, regression testing, protocol and packet validation, hardware-in-the-loop and bench testing, failure reproduction, instrumentation, logging, CI-aware test workflows, release validation, and root-cause analysis.
- Vehicle, IoT, and telematics systems: ECU/tuning workflows, cellular/GPS/connected-device concepts, BLE/CAN/OBD2 telemetry, raw frame capture, signal decoding, thousands of logged parameters, derived metrics, fuel/sensor controllers, and embedded Linux/networking concepts.
- Scientific and regulated background: biochemistry, analytical testing software, medical/technical device-adjacent code, data integrity, traceable workflows, documentation, validation, and quality-focused change control.
- Working style: ask scope questions early, understand the user, keep changes readable, document decisions, review AI-generated code critically, communicate before going dark, and own production behavior.

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Professional Experience

Stealth Machine Tools / Stealth Batteries

Software Engineer / AI Engineer / Technical Lead

May 2025 - Present

- Lead and mentor three junior developers while building full-stack software, AI integrations, CRM systems, support automation, dashboards, field/support apps, and internal tools for Stealth Machine Tools, Stealth Batteries, and related businesses.
- Architected and developed full-stack CAD/CAM, operator, configuration, diagnostics, and support interfaces for fiber laser welders and cutters, plasma CNC machines, press brakes, benders, and related industrial equipment.
- Built cross-platform Flutter/Dart machine applications for Windows, macOS, iOS, and Android, connecting user interfaces and cloud services to machine settings, telemetry, logs, diagnostics, firmware, customer records, and remote-support workflows.
- Worked across embedded controllers, microcontrollers, sensors, machine communications, firmware behavior, and application-to-device integration; developed Python tooling for automated validation, repeatable diagnostics, regression testing, log analysis, and production troubleshooting.
- Supported firmware and hardware testing with UART/serial, TCP/IP, BLE, embedded Linux and microcontroller development platforms, and laboratory tools including oscilloscopes, multimeters, logic analyzers, and bench power supplies.
- Built AI-assisted customer support and internal workflow systems with customer chat, onboarding, claim/warranty intake, document/context lookup, ticketing, logging, analytics, escalation paths, and human review workflows.
- Built Oracle APEX-backed customer, warranty, automation, and support databases for Stealth Machine Tools and Burger Motorsports, turning business requirements into database applications instead of manual spreadsheets and inbox tracking.
- Built Stealth Batteries commerce/admin platform using Next.js, React, TypeScript, Payload CMS, Postgres, Stripe, webhooks, email providers, shipping APIs, dealer portals, affiliate workflows, sales-rep dashboards, order management, warranty/support workflows, and custom admin tools.
- Built Stealth Machine backend tooling with Next.js/TypeScript/Prisma patterns around auth, customers, machines, tickets, training/support materials, AI configuration, analytics, and admin workflows.
- Built cross-platform Flutter field/support tooling for machine support and training workflows, including API integration, local storage, auth helpers, camera/speech/PDF support, QR-style workflows, charts, and field-service user experience.
- Built CRM/order/customer tracking systems with Node/Express, SQLite, JWT auth, role permissions, admin approvals, customer portals, document intake, real-time updates, Twilio/Nodemailer notifications, audit trails, deployment notes, and Oracle APEX-connected operational records.
- Connected customer-facing portals, admin work, order state, approvals, document uploads, notifications, reporting, support history, warranty claims, shipping/return steps, and customer records into structured workflows that reduced manual operational load.
- Built support workflows that could register customers, collect product/warranty information, track claim status, assist with returns, create shipping/return workflows, and reduce repetitive staff work.
- Worked across ecommerce, industrial equipment support, customer onboarding, warranty/claims handling, inventory visibility, admin dashboards, operational reporting, and AI-assisted employee workflows.
- Used AI development tools daily for implementation, code review, documentation, refactoring, debugging, and workflow design while retaining responsibility for correctness, security, and production behavior.

Burger Motorsports

Lead Developer - Automotive Software, Mobile Apps, Hardware Integration, Support Platforms

2019 - 2025

- Served as principal vehicle software engineer and lead developer across the complete JB4 product stack: embedded firmware, bootloaders, tuning logic, connected hardware, cross-platform user interfaces, backend/support systems, databases, and production validation for an ecosystem used by more than 500,000 vehicles worldwide.
- Coordinated work across software, hardware, firmware-adjacent requirements, vehicle platform behavior, support needs, customer usability, release risk, and product documentation.
- Built the full-stack interface around JB4 tuning hardware, including Flutter/Dart applications for iOS, Android, Windows, and macOS; BLE/serial device communication; cloud and support services; Oracle APEX operational databases; and customer, tuner, and developer interfaces.
- Developed and supported embedded C/C++ firmware and microcontroller integrations for JB4 devices, CANFlex controllers, standalone GDI drivers, fuel injector controllers, sensor modules, and related automotive hardware.
- Built vehicle tuning, calibration, diagnostics, and datalogging systems capable of collecting, decoding, visualizing, and exporting thousands of parameters across many vehicle platforms, ECUs, firmware versions, maps, and hardware configurations.
- Implemented remote application-to-device firmware delivery including firmware retrieval and manual HEX loading, bootloader entry, erase/write commands, block flashing, checksums, version validation, progress state, recovery handling, bootloader exit, and user-facing safety controls.
- Built automated and bench validation workflows using Python and engineering tools to test firmware, protocols, telemetry, connection recovery, regression behavior, and hardware/software interaction before release.
- Built command/protocol logic for JB4 hardware including settings retrieval, settings save, logging start/pause, map selection, E85/flex fuel configuration, WMI/meth settings, boost/fuel tables, AFR/meth/fuel pressure telemetry, and developer packet inspection.
- Developed CANFlex/NewCANFlex mobile hardware software for fuel and sensor control: BLE device communication, ethanol content, fuel temperature, fuel pressure, RPM, speed, water/oil temperature, gear and torque surfaces, CAN bus output configuration, analog output modes, pressure sensor modes, calibration controls, firmware/version display, live gauges, logs, and ECU/JB4 integration paths.
- Built CRM-backed support tooling that replaced manual email-based log review with structured customer submissions, vehicle log intake, AI-assisted parsing, diagnostic/tuning guidance, support history, escalation paths, and Oracle APEX-backed operational records.
- Helped reduce operational support burden by creating systems that allowed technical support workflows, product context, vehicle logs, and customer history to live in structured tools instead of inboxes and spreadsheets.
- Worked close to embedded systems, custom hardware, firmware, and vehicle communication protocols, translating raw device/vehicle data into reliable UI state and supportable diagnostic workflows.
- Built across desktop and mobile hardware interfaces, including legacy C# WinForms/serial/BLE support code and newer Flutter mobile tooling.
- Operated in a domain where incorrect parsing, settings, firmware assumptions, or support guidance could affect real vehicles, requiring careful validation, documentation, ownership, and code review.
- Completed the major software initiatives ahead of schedule, documented and transitioned the systems, and moved on once the core work had been delivered and the role naturally wound down.

Independent Vehicle / Hardware Software Projects

Vehicle Telemetry, CAN Integration, Firmware-Adjacent Tools

- Built TurboLamik AWD/transmission telemetry foundation with ESP-IDF-oriented firmware scaffolding, passive CAN listening on a shared 500 kbps bus, selected BMW E90 CAN ID decoding, TurboLamik TCU frame decoding, normalized drivetrain/chassis signals, derived metrics, health/watchdog state, raw CAN capture, and shadow-mode AWD request logic.
- Reverse engineered TurboLamik ADX protocol behavior into a client-device spec, including serial assumptions, polling sequence, fixed packet maps, gearbox torque, engine torque, input/output RPM, slip/clutch data, linear pressure, lockup, analog inputs, CAN torque fields, AWD activation, PWM output, steering angle, and error/safe-mode data.
- Built BMW E90 PT-CAN-style signal maps for throttle, engine RPM, wheel speeds, coolant/oil temperature, steering angle, brake proxy, handbrake, reverse, and validation notes separating confirmed decodes from provisional in-car validation items.
- Documented MaxxECU default CAN output integration for 11-bit IDs at 500 kbit/s, including RPM, throttle, MAP, lambda, ignition, fuel duty, vehicle speed, slip, traction flags, brake/clutch flags, accelerations, oil pressure/temp, boost target, transmission temp, differential temp, and firmware-version-gated protocol differences.
- Designed Flutter telemetry/logging app shells for vehicle-side diagnostics, including device status, live dashboard, raw frame monitor, decoded signal monitor, capture control, saved sessions/export, and vehicle profile display.
- Built AWD/transfer-case tuning application concepts with Flutter domain models for torque estimates, slip resolution, predictive layers, drive rules, PWM commands, map interpolation, profiles, real-time diagnostics, configuration pages, and BLE sync paths.
- Built ESP32 and touch-input hardware projects demonstrating compact embedded input handling, USB HID-style control, driver integration, Python GUI/control scripts, and physical-device testing workflows.
- Developed across a broad range of microcontrollers and embedded platforms, including Arduino and Raspberry Pi-class systems, with UART/serial communications, TCP/IP networking, embedded Linux, BLE, CAN, GPIO, PWM, persistent configuration, and device-level troubleshooting.
- Used oscilloscopes, logic analyzers, digital multimeters, bench power supplies, and serial/CAN diagnostic equipment to validate signals, power behavior, timing, communications, firmware changes, and failure conditions.
- Built Python-based automated test and data-analysis tools for repeatable device validation, packet/protocol testing, regression coverage, telemetry inspection, and production issue reproduction, with CI-oriented workflows where appropriate.

Bayer

Technical / Analytical Testing Software Background

2015 - 2018

- Worked in analytical/scientific environments after studying biochemistry and software engineering.
- Built and supported custom code, analytical workflows, and technical operations for regulated testing environments.
- Gained experience with documentation, data integrity, regulated processes, quality expectations, validation, troubleshooting, careful change control, and technical systems where small mistakes could affect real outcomes.
- Developed a strong foundation in scientific thinking, root-cause analysis, measurement workflows, traceability, and software that supports technical users instead of just looking polished.

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Selected Project Highlights

Tech Support AI Backend

Backend AI support system built around real operations: Claude integration, JWT auth, MongoDB/Mongoose models, machine/customer records, support tickets, document ingestion, analytics, notifications, and routes that turn a customer conversation into a trackable backend workflow.

Stealth AI Support System

Customer and internal support system with Claude-style chat, real-time support workflows, admin dashboard concepts, ticketing, analytics, conversation history, escalation, warranty/claims handling, onboarding, and human handoff paths.

Stealth Batteries Commerce/Admin Platform

Production business platform connecting ecommerce, products, dealers, affiliates, sales reps, orders, shipping, support, warranty, dashboards, and Payload/Postgres admin tooling. The system tied customer-facing workflows to internal operations so the business could move faster without relying on manual email/status tracking.

Stealth Machine Backend and Field App

Next.js/Prisma backend plus Flutter field/support app for machine records, support tickets, training materials, customer context, AI support configuration, analytics, local storage, camera/speech/PDF helpers, QR-style workflows, charts, and field-service support UX.

Industrial Machine CAD/CAM and Control Software

Full-stack CAD/CAM, operator, configuration, diagnostics, telemetry, and remote-support software for fiber laser welders and cutters, plasma CNC machines, press brakes, benders, and related manufacturing equipment, spanning cross-platform Flutter/Dart interfaces, backend services, databases, machine communications, embedded controllers, firmware workflows, and field validation.

Customer Tracking CRM

CRM/order workflow software for companies, contacts, orders, admin approvals, customer portal views, document uploads, OCR-style processing, real-time updates, notifications, audit trails, role-aware workflows, and Oracle APEX-connected database operations.

GEO Command Center Platform

Full-stack AI/search visibility SaaS with Next.js frontend, Supabase/Postgres/Drizzle data modeling, FastAPI backend services, worker/background processing, tests, provider/report pipelines, audits, citation tracking, multi-tenant dashboards, and analytics workflows.

JB4Pro Vehicle Tuning Platform

End-to-end embedded and full-stack platform for JB4 tuning hardware, including C/C++ firmware, microcontrollers, bootloaders, remote application-to-device firmware delivery, BLE/serial/CAN communication, Flutter applications across mobile and desktop, vehicle calibration, live telemetry, logging of thousands of parameters, settings and map control, E85/flex fuel and WMI/meth workflows, diagnostics, cloud/support systems, and production validation.

CANFlex Fuel and Sensor Controller

Mobile software for a fuel injector / E85 sensor / fuel pressure control ecosystem with BLE communication, ethanol/fuel-temperature/fuel-pressure gauges, CAN bus and analog output configuration, pressure sensor modes, calibration, logs, firmware/version handling, and ECU/JB4 integration paths.

TurboLamik AWD / Transmission Telemetry

Vehicle/embedded integration project with ESP-IDF-oriented C scaffold, BMW E90 CAN signal decoding, TurboLamik TCU frame decoding, raw CAN capture, health/watchdog state, derived drivetrain metrics, shadow AWD request logic, BLE telemetry, Flutter dashboard, raw frame monitor, decoded signal monitor, and saved log/export workflows.

STR8Tune ECU Calibration Editor

ECU calibration/editor tooling with React and TypeScript: XDF/BIN parsing concepts, editable tables/maps, hex and calibration views, tuning assistant flows, comparison tools, 2D/3D visualization, and export-style workflows.

AmpGen Configurator

Next.js/React product configurator for electrical/power-system planning, turning a physical product setup into an interactive customer/operations tool with draggable layout zones and TypeScript UI logic.

Industrial Commerce and Admin Systems

Built commerce/admin patterns for industrial and specialty product businesses, including inventory services, manual order flows, checkout/payment logic, shipping dashboards, customer/admin workflows, migrations, reusable components, API documentation, and handoff notes.

Windows and Network Automation

Built practical Python/PowerShell local API and Windows/network automation tooling around hotspot/control workflows, showing the operational scripting side of keeping real systems usable.

Legacy Hardware Desktop Interfaces

Built and maintained C# WinForms-era hardware interface code around serial port detection, Bluetooth/BLE support, Windows UI surfaces, device discovery, and tuning-tool ergonomics.

Technologies

Languages: JavaScript, TypeScript, Python, SQL, Dart, C#, C++, C, Java, HTML, CSS, Bash, PowerShell, VBA

Frontend: React, Next.js, Tailwind CSS, Radix/shadcn-style UI, dashboard UI, admin panels, customer portals, configuration interfaces, charts/gauges, table/map editors, 2D/3D visualization concepts

Backend: Node.js, Express, FastAPI, REST APIs, WebSockets, JWT, OAuth concepts, role permissions, audit logging, provider adapters, webhooks, background jobs, notification workflows

Data: Oracle APEX, SQL, PL/SQL-style workflows, Postgres, SQLite, MongoDB/Mongoose, Supabase, Drizzle, Prisma, Redis/BullMQ-style queues, Payload CMS, data modeling, migrations, reporting tables, CRM/order state

AI / Integrations: Claude/Anthropic, OpenAI-style APIs, AI support agents, prompt/context workflows, document ingestion, OCR/document processing, ticket triage, analytics, Stripe, shipping APIs, UPS workflows, Resend/Nodemailer, Twilio, webhooks

Industrial / CAD-CAM: fiber laser welders and cutters, plasma CNC systems, press brakes, benders, operator HMIs, CAD/CAM workflows, machine configuration, diagnostics, telemetry, remote support, and field-service software

Mobile / Hardware: Flutter, Dart, iOS, Android, Windows, macOS, BLE, UART/serial, CAN bus, OBD2, TCP/IP, cellular/GPS/telematics concepts, raw frame capture, diagnostics, high-volume datalogging, gauges, calibration, and application-to-device control

Embedded / Firmware: C, C++, microcontrollers, ESP32, ESP-IDF, Arduino, Raspberry Pi, embedded Linux, GPIO, PWM, interrupts, drivers, watchdogs, state machines, persistent configuration, bootloaders, remote firmware updates, HEX parsing, erase/write, checksums, and recovery flows

Validation / Test: Python automation, firmware validation, regression testing, protocol and packet tests, hardware-in-the-loop and bench testing, CI pipelines, automated test frameworks, logging, failure reproduction, root-cause analysis, oscilloscopes, logic analyzers, multimeters, power supplies, and serial/CAN tools

Vehicle / Telematics: ECU calibration and tuning, JB4, CANFlex, GDI drivers, CAN RX dispatch, signal stores, thousands of telemetry parameters, BMW E90 decoding, TurboLamik TCU decoding, MaxxECU CAN integration, fuel/sensor controller workflows, and connected-device diagnostics

Desktop / Ops: C# WinForms, serial port monitoring, Python GUI tools, PowerShell scripts, Windows/server automation, Docker Compose, Git, GitHub, CI/CD, and technical documentation

Education

Biochemistry and Software Engineering coursework, 2012 - 2015

Studied biochemistry with a software engineering focus while preparing to build analytical testing and technical workflow software.

How I Work

I am comfortable with "here is the problem, go figure it out" work. I ask scope questions early, especially around users, safety, diagnostics, edge cases, ownership, and production risk. I prefer simple architecture, clear data flow, validation before control output, and code that another engineer can safely change. I communicate before going dark, leave notes/change logs, and document enough context for the next engineer or AI-assisted workflow to understand the system later.